

Assignment 3

Total: 5 marks

This assignment must be hand-written. **Show ALL steps in ALL questions.**

Read the questions carefully.

1. Design a 3 bit parallel adder & subtractor using full adders. But, you only have XOR gates and some decoders & encoders. [2]

2. Design $F(A,B,C,D) = \sum(1,3,4,5,9,13,14,15)$ using only one 4:1 MUX and basic gates. [1]

3. Design a 4 bit parallel adder that will do the following: [2]

If either input X is greater than or equal 8 and divisible by 4 or Y is greater than 7, Perform X-Y

Else: X+Y